Project Develpoment Phase

Date	8 July 2026
Team ID	
Project Name	Food Ordering And Customer Trends
Maximum Marks	4 Marks

1. Data Preprocessing Steps

Step 1: Import the Dataset

- Imported the **Food Ordering Behaviour and Consumer Trends** dataset into Tableau Public.
- Verified that all 50,000 records and operational fields (Count of food_orders, Age Group, Cuisine, Meal Type, City, Order Value, Delivery Fee, Avg. Rating, Restaurant Type, Company Type, Mood, etc.) were imported successfully.
- Confirmed that the database structure was ready for active preprocessing.

Step 2: Check for Missing Values

- Examined the entire dataset for null, blank, or missing values across all structural attributes.
- Verified that critical transactional metrics such as Order Value, Delivery Fee, Customer Rating, Cuisine, Meal Type, and City contained fully populated, valid information.
- Handled any anomalies appropriately to prevent downstream errors during visual analysis.

Step 3: Remove Duplicate Records

- Checked the dataset for duplicate order records using unique identifier metrics.
- Removed duplicate entries to prevent artificial inflation of metrics and inaccurate counts.
- Ensured that every distinct transaction was represented exactly once, preserving th true scale of the data (50,000 distinct records).

Step 4: Validate Data Types

Ensured that each attribute within the cleaned dataset was mapped to its correct structural data type for optimized filtering and visualization:

- **Count of food_orders** → Number (Integer)
- **Order Value** → Number (Decimal/Currency)
- **Delivery Fee** → Number (Decimal/Currency)
- **Avg. Rating** → Number (Decimal)
- **Avg. Time Taken To Order** → Number (Continuous)
- **Customer Age Group** → Categorical
- **Cuisine** → Categorical
- **Meal Type** → Categorical
- **Restaurant Type** → Categorical
- **Company Type** → Categorical
- **Mood** → Categorical
- **City** → Geographic / String

Step 5: Verify Data Consistency

- Scanned all rows for unrealistic or invalid values (such as negative order values, out-of-bounds ratings, or empty string errors).
- Standardized text inputs across categorical variables, ensuring consistent naming

formatting for cities, specific cuisines, and meal types.
- Verified logic boundaries across structural dimensions before proceeding to building the final charts.

Step 6: Prepare Calculated Fields & Aggregations

Created custom calculations, operational metrics, and explicit performance definitions required for the dashboard execution:

- Count of food_orders (Total dynamic order tracking)
- Total Delivery Fee (Summed logistical costs)
- Total Order Value (Gross transactional volume)
- Avg. Rating (Overall customer satisfaction level)
- Avg. Time Taken To Order (Mood-based time metrics)

Step 7: Final Clean Dataset Summary

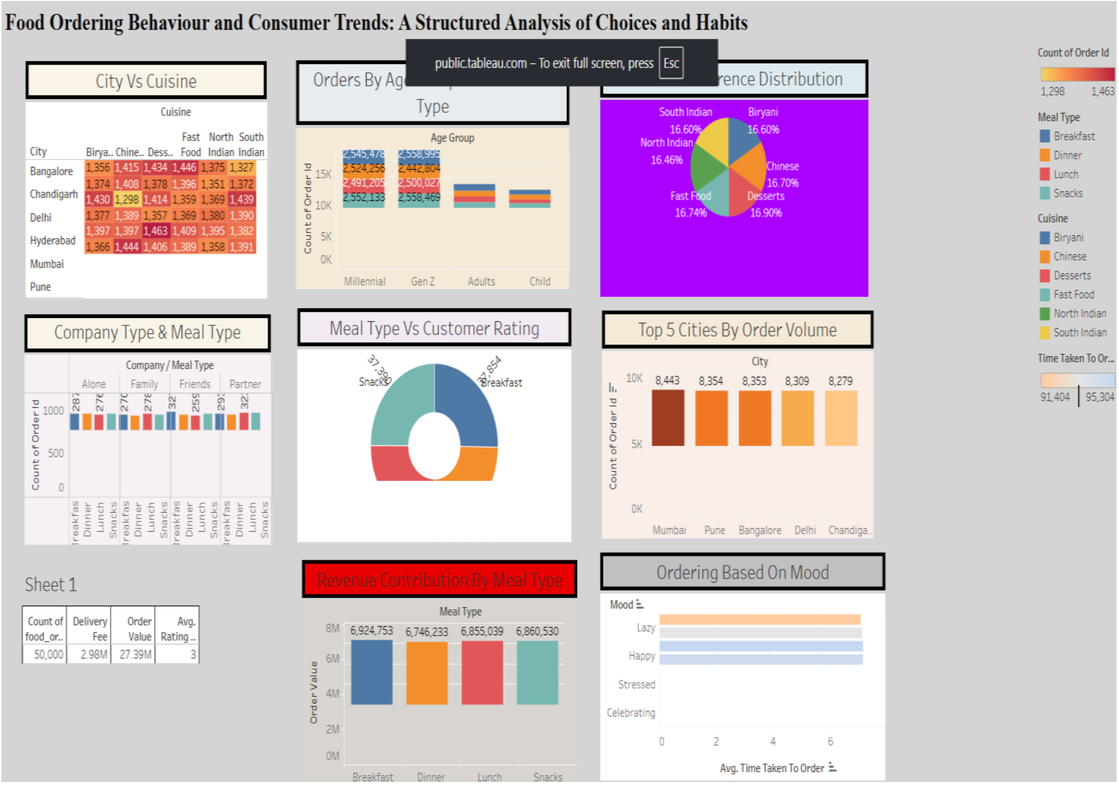
- Completed validation checks to ensure the preprocessed dataset was accurate, structurally integrated, and completely ready for dashboard implementation.
- The finalized operational dataset was successfully loaded directly into the primary sheets of the Tableau Public workspace environment.

2. Business Questions with Implemented Visualization Matrix

S.No.	**Business Question**	**Implemented Visualization**
1	What are the total summary values, t volumes, logistical distribution fees, satisfaction averages after data prepr	**KPI Demonstration Matrix (She** *(Summary Card Table displaying 5 2.98M Delivery Fees, 27.39M Ord an Avg. Rating of 3)*
2	How do specific cuisine types compa when analyzed across different geogr distribution hubs?	**City Vs Cuisine** *(Cross-tabulated Heatmap showin order distribution densities)*
3	How do ordering habits split across c customer age categories based on the of the restaurant segment?	**Orders By Age Group & Restaur** *(Segmented Stacked Bar Chart ma Millennial, Gen Z, Adults, and Chi*
4	What is the total proportional breakd volume distribution of consumer pref core food groups?	**Cuisine Preference Distribution** *(Color-coded Pie Chart mapping S Biryani, Fast Food, and Desserts)*

S.No.	Business Question	Implemented Visualization
5	What specific volume interaction pat when mapping social dining contexts meal periods?	**Company Type & Meal Type Or** *(Grouped Multi-Variable Bar Char Alone, Family, Friends, and Partn*
6	How are customer satisfaction rating across the major operational meal pe day?	**Meal Type Vs Customer Rating** *(Centered Donut Chart analyzing s satisfaction levels)*
7	Which specific metropolitan regions single highest baseline of consumer c transaction frequency?	**Top 5 Cities By Order Volume** *(Rank-sorted Descending Bar Cha Mumbai, Pune, Bangalore, Delhi, Chandigarh)*
8	How do daily dining financial values directly with customer emotional stat ordering process durations?	**Revenue Contribution By Meal T Ordering Based On Mood** *(Dual-View Combo Visual featurin Bar Chart and a Mood-based Aver Horizontal Bar Chart)*

Dashboard :



Story :